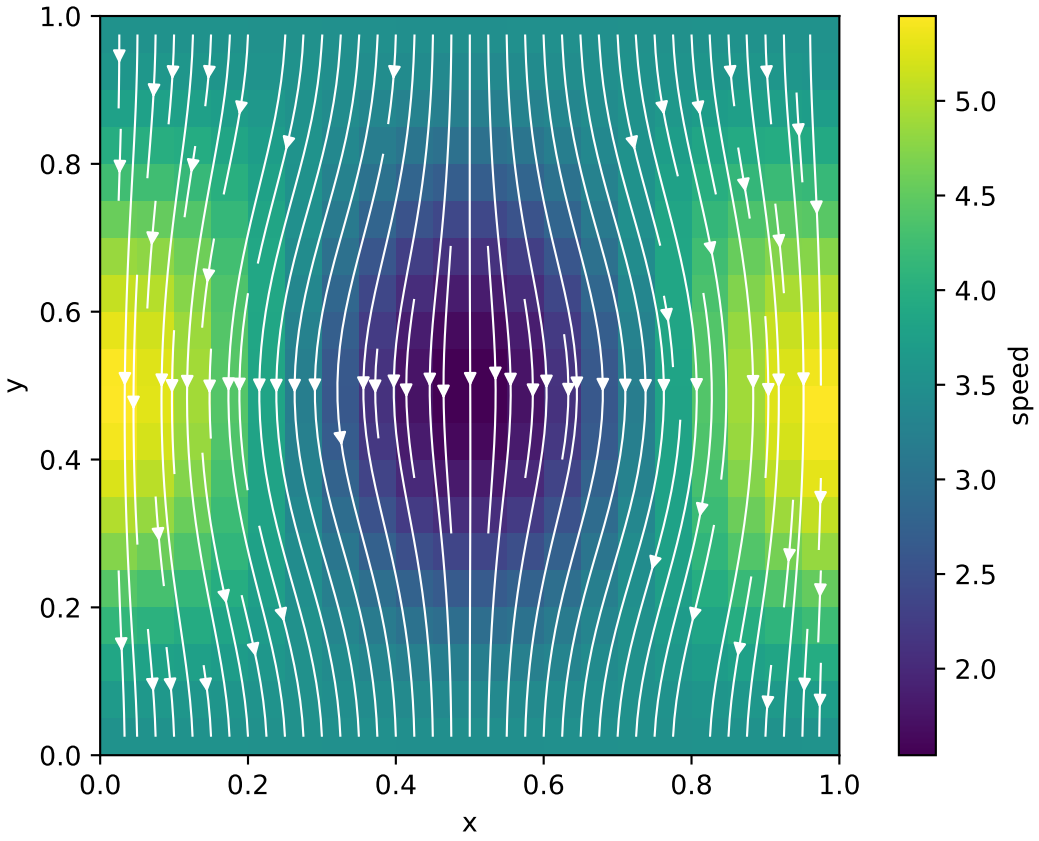
Sean Gravelle
Numerical PDEs Homework #1

Problem 1: Solve Stokes

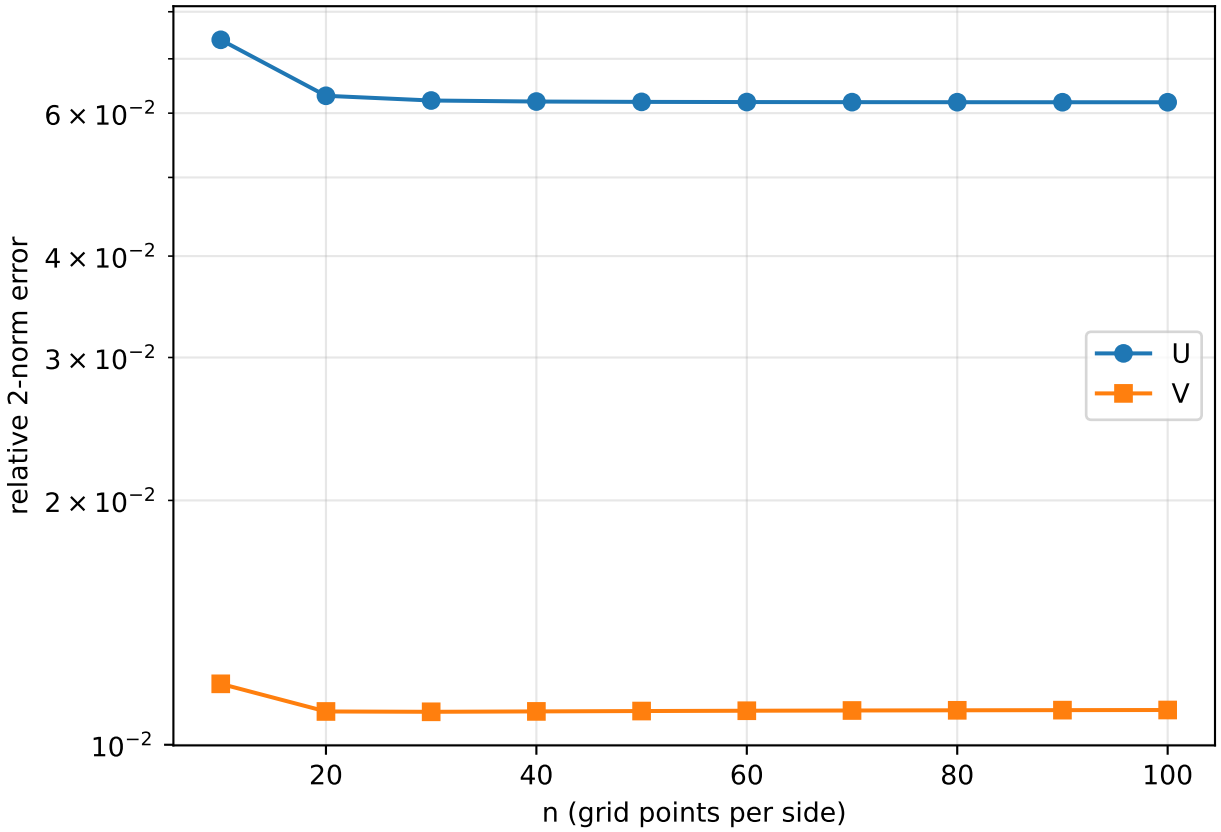
My solution to the Stokes equations in 2 dimensions, with the given boundary conditions, is plotted below. Arrows indicate direction of the fluid's flow and color indicates the magnitude of the solution vector; that is, the speed of the fluid.



Problem 2: Error Analysis

Plotting the relative error of numerical solutions on an $n \times n$ grid gives the following errors for U and V. The error for P also was essentially level, but on the order of 1 (instead of 0.01) so I excluded it from this graph. This occurred even though I did subtract off the mean of my numerical solution for P.

While error on the order of 0.01 is somewhat small, the fact that it is level, not decreasing, as a function of n is unexpected.



AI disclosure statement: Both plots were made with LLM assistance. All mathematical work and all code to produce the numerical solutions and error calculations are my own, without LLM.